

Use Case Pay Order by Credit Card

1. Use Case Code

UC003

2. Brief Description

The use case “Pay Order by Credit Card” describes the interaction between the Customer, the AIMS Software, and the PayPal payment gateway. It is invoked when the Customer chooses to pay for their order using a credit card rather than the default QR code method.

3. Actors

There are two actors:

Primary Actor: Customer.

Secondary Actor: PayPal (Payment Gateway).

4. Preconditions

- + The use case “Place Order” has been initiated.
- + The Customer has confirmed the invoice information.

- + The total amount, including VAT and shipping fees, has been calculated.
- + The Customer has selected "Credit Card" as the payment method.

5. Basic Flow of Events

1. AIMS Software displays the credit card information entry form.
2. Customer enters credit card details (Card Number, Holder Name, Expiration Date, and CVV) and submits.
3. AIMS Software validates the format of the entered data.
4. AIMS Software sends a payment request with the transaction amount to the PayPal system.
5. PayPal authenticates the card and checks for sufficient balance.
6. PayPal processes the transaction and returns a "Successful" status and transaction ID to AIMS Software.
7. AIMS Software records the payment transaction details.

8. AIMS Software displays general information of the order (customer name, phone number, shipping address, province, total amount) and transaction information (transaction ID, transaction content and transaction datetime).
9. AIMS Software sends invoice and payment transaction information to the customer's email.

6. Alternative Flows

No	Location	Condition	Action	Resume Location
1	Step 3	Invalid card data format	AIMS Software notifies the user of the specific error	Step 1
2	Step 5	Authentication failed or Insufficient balance	AIMS Software displays a Failed message.	Use case ends

3	Step 4	Connection timeout with PayPal	AIMS Software notifies that the payment gateway is unavailable.	Use case ends
4	Step 1	Customer cancels payment	AIMS Software returns to the payment method selection screen.	Use case ends

7. Input Data

No	Data Fields	Description	Mandatory	Valid Condition	Example
1	Card Number	The 16-digit credit card number	Yes	Must be numeric	2560 4508 3456 6789
2	Cardholder Name	Name as it appears on the card	Yes	Alphabetic characters only	PHAM DUC PHUOC
3	Expiration Date	The card's expiry date	Yes	MM/YY format and must	03/30

				be in the future	
4	CVV/CVC	Security code on the back	Yes	Exactly 3 digits	222

8. Output Data

No	Data Fields	Description	Display Format	Example
1	Transaction ID	Unique ID provided by PayPal	Alphanumeric	PDP-12345678
2	Amount	Final charged amount	Integer with comma separator	3,636,366 VNĐ
3	Date	Date and time of transaction	dd/mm/yyyy hh:mm	03/06/2025 03:06

4	Status	Outcome of the process	Text	Successful
5	Transaction Content	Purpose of transaction	Text	Payment for order #98765

9. Postconditions

+ Success: A new transaction record is saved, the order status is updated to " Pending processing state," and the cart is emptied.

+ Failure: No payment is recorded, and the order remains unpaid.